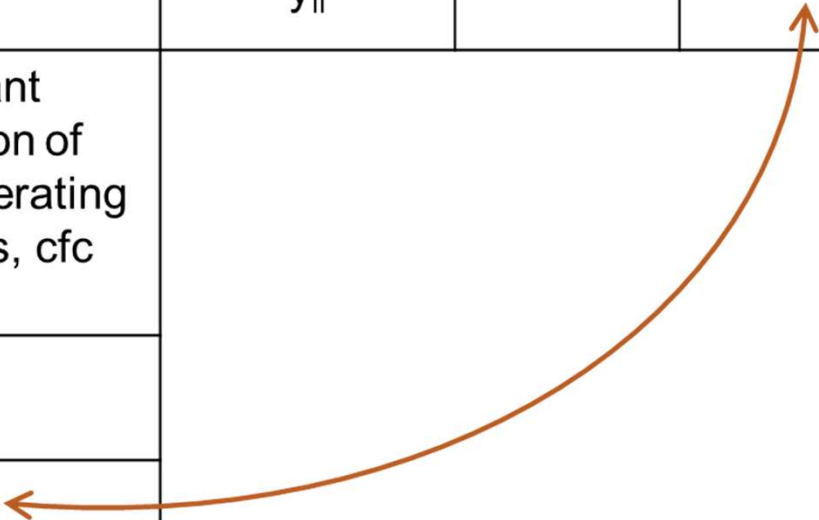


OPERATIONS RESEARCH AND SYSTEMS ANALYSIS

Tutorial: Input-output analysis - basics

Input-output model

- Each element/industry act as supplier and consumer/user.

		Industries/elements as users/consumers	Final uses	Final uses total	Used resources total/ <i>total outputs</i>
		1, 2, ... n	1, 2, ... h		
Industries/elements as suppliers	1 2 : n	1 st quadrant Intermediate consumption x_{ij}	2 nd quadrant Final uses y_{il}	Y_i	X_i
Value added	1 2 : n	3 rd quadrant Compensation of employees, operating surplus, taxes, cfc z_{kj}			
Total output Import					
Total supplies/ <i>inputs</i>		X_j			

Example I

Millions CZK	Primary sector	Secondary sector	Tertiary sector	Final uses	Total Outputs
Primary sector	22 125	319 949	...	122 696	...
Secondary sector	76 477	...	574 790	3 656 466	...
Tertiary sector	47 492	702 888	...	2 434 413	4 666 760
Wages and salaries	39 516	403 538	765 936		
Income	46 080	...	658 613		
Other inputs	...	400 511	890 034		
Import	234 009	2 011 283	277 641		
Total Inputs	...	6 631 790	...		

Example I tasks

1. Fill the missing values
2. Construct the coefficient matrix A
3. Construct the coefficient matrix M
4. It is possible to repeal the limits for the coal mining. This would result in the 5% growth of total outputs from primary sector. What final uses (final production) could be expected? Compare the new values with the original from table 1.

Example I tasks

5. The government decided to cut the expenditures and reduces the final uses in secondary sector by 5% and by 10% in tertiary sector. What would be the impact in the total output? Compare the new values with the original from table 1.

Example II: DIY

Small focused on personal finance counseling is described by the following IO table. The main activities consist of finance counseling, marketing and project management.

	Finance	Marketing	PM	Y	X
Finance	3	4	...	...	45
Marketing	...	2	10	3	...
PM	10	8	...	5	...
Wages	10	5	5		
Inputs	3	...	6		
EBIT	9	-2	0		
X	...	...	25		

Example II: DIY

1. Fill the missing values
2. Construct the coefficient matrix **A**
3. The company has the possibility to expand and as a result the total output of finance counseling would rise to 55 mil EUR.
 - a. What final uses (final production) could be expected in case that the total output of other activities will not change?
 - b. What will be the new EBIT?

Example II: DIY



4. Management of the company plans to offer more project management services to their clients. The required final output of PM is 10 millions EUR.
 - a. What is the necessary increase of total outputs?
 - b. What will be the expected EBIT?

Thank you for your attention

